

PurityLoop AI

MRF Operations and Frontend Design Audit

Navigation stability, operational usefulness, and design-quality review

AUDIT OUTCOME

Critical navigation defects confirmed. No functional workflow changes recommended before route-shell and initialisation stabilisation.

Executive summary

PurityLoop has the right workflow pieces, but it does not yet feel dependable enough for an MRF control desk. The primary risk is technical: route-specific styling and page initialisation are unstable, allowing wrong or missing page treatment during navigation. That undermines trust in numbers and warnings.

The design is also too dashboard-decorative: repeated green surfaces, equally weighted cards, weak warning contrast, and titles that describe UI rather than operational decisions.

Reproduction and root-cause analysis

Critical - Route style ownership is broken

The rendered analytics shell has `data-page=analytics`, but body ends with no class and no `data-page`. `PageHtml` imperatively writes route state to React-owned body; layout owns body without those attributes. Many late rules depend on `body[data-page]`. Impact: missing route treatment, default-size charts, visual jump, and possible wrong-page styling during replacement.

High - Every route initialises every feature

The shared script refreshes data then runs upload, result, review-modal, analytics, and drill-through initialisers on all routes, at DOM ready and custom page-ready. Destination markup appears before data/charts settle, creating timing-dependent layout changes and stale-state risk in production.

Medium - Motion targets the unstable body

The global loading fade/translate is applied to body and removed late, creating a whole-page settle rather than a stable local transition.

Test condition

No browser console errors appeared. The local frontend points to a remote ngrok API, so data-dependent pages remained in genuine empty/loading states. This supports layout and state review, not populated-value validation.

Dashboard / Operations overview

OVERALL DESIGN SCORE

4/10

What works

- Core conditions are present: review queue, confidence, recoverable value, contamination, latest upload.
- Empty state does not invent production figures.

Design problems

- Critical - Page-specific style can disappear after hydration, so cards and charts lose intended dimensions. Fix route-shell selector ownership now.
- High - Needs Review does not dominate the five-second read. Make it first, amber/red, with a direct Open queue action.
- High - Five insight tiles create scanning work with no decision order. Keep pending review, contamination risk, recovery value, last batch.
- Medium - Analytics Overview is generic. Rename it Operations Dashboard: Exceptions, quality, and recovery value.
- Medium - Repeated green surfaces flatten hierarchy. Green should mean confirmed/normal, not default decoration.
- Medium - Empty charts are over-sized. Collapse to No scans in selected period plus Upload images.

MRF manager reaction

- I need to know whether a hazardous or low-confidence batch needs intervention. Instead I read four equal cards, five tiny insights, then charts.

Recommended changes

- Implement route-style stability now. Promote exceptions and last-batch condition. Keep drill-through; reserve KPI restructuring for redesign.

Analytics

OVERALL DESIGN SCORE

5/10

What works

- Date range is appropriate. Material mix, value by category, and scan trend are sensible chart choices. Drill-through is worth preserving.

Design problems

- Critical - Charts render before stable dimensions and resize after data arrives.
- High - Recoverable Value needs RM in title/axis, not implication only.
- High - Trend lacks period, baseline, and operational interpretation.
- Medium - Green-led chart treatment fails to distinguish normal, review, and contamination.
- Medium - Manager Actions competes with charts; when non-empty it should sit above analysis.
- Low - Legends and axes are too small for shift-length use.

MRF manager reaction

- Drill-through is valuable, but I should not hunt below charts to learn what needs action now.

Recommended changes

- Stabilise containers now. Put Manager Actions first when actionable. Use red for hazardous, amber for review, green for confirmed recyclable material. Add one-line chart insights.

Upload

OVERALL DESIGN SCORE

6/10

What works

- Upload is correctly the first workflow action. Batch and ZIP processing are represented.

Design problems

- High - Formats, batch limit, ZIP extraction, and recovery from a bad file need a concise rules block at the dropzone.
- High - Queue/progress region must reserve final height to prevent controls moving while processing.
- Medium - Success needs an explicit next action: Review result or Upload another batch.
- Medium - Dashboard-style card chrome makes a task screen feel heavier than needed.
- Low - Green emphasis is overused before upload succeeds.

MRF manager reaction

- Can I upload a line-camera ZIP? How many images? What happens to a corrupt file? I should know before selecting files.

Recommended changes

- Fixed-height queue, per-file retry/failure, direct next action. Implement reserved state space now; redesign density later.

Result

OVERALL DESIGN SCORE

5/10

What works

- Detection, confidence, material facts, and human review are present. Review controls are close to detection outcome.

Design problems

- High - Reading order is unclear. It should be hazard/review status, detected material, confidence, action, then estimates.
- High - Recommended action must outrank estimated value. A battery/contaminant cannot look secondary.
- Medium - Weight/value need Estimated from class model labeling to avoid false precision.
- Medium - Repeated cards turn one scan into a dashboard rather than a focused workbench.
- Low - Page title should identify the batch/item, not only Result.

MRF manager reaction

- Tell me whether to stop, quarantine, confirm, or pass. Do not make me compare cards.

Recommended changes

- Keep review workflow. Give action a persistent high-contrast panel; group estimates beneath it. Reserve broader workbench restructuring for redesign.

Log / Verification history

OVERALL DESIGN SCORE	5/10
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What works

- Logs, filters, status, and review history are in the right functional area. A table is appropriate for an audit trail.

Design problems

- High - Small dense text and pills weaken scanning. Timestamp, material, confidence, status, reviewer, exception reason need a fixed hierarchy.
- High - Exception rows need a distinct marker and direct Review action.
- Medium - Filters need active scope and result count.
- Medium - History is weaker than Verification Log or Review Queue.
- Low - Large table containers add empty space without improving scanability.

MRF manager reaction

- I need unresolved exceptions in seconds, then who cleared them and why.

Recommended changes

- Default to unresolved exceptions; pin action/status, use tabular figures, highlight hazardous/review rows, retain full history.

Review modal

OVERALL DESIGN SCORE

6/10

What works

- Modal is fixed within viewport with an internal scroll region; that is correct.

Design problems

- High - Focus hierarchy must be evidence image, AI decision, operator correction, confirmation.
- High - Confirm and Reject/Quarantine need unmistakable labels and unequal risk treatment.
- Medium - Blur/dark treatment is decorative unless foreground contrast is materially stronger.
- Medium - No unsaved-change safeguard established.
- Low - Older history should collapse behind latest decision.

MRF manager reaction

- One wrong click becomes a bad report. I need no ambiguity about what I approve.

Recommended changes

- Keep viewport behaviour. One primary confirmation, separated quarantine/reject, explicit cancel. Add focus management and unsaved-change protection later.

Drill-through analysis

OVERALL DESIGN SCORE

6/10

What works

- Important detail workflow exists and must remain. It connects overview to material evidence.

Design problems

- High - It sits too far below overview and competes with many panels.
- Medium - Panels repeat labels/cards instead of answering why a KPI changed.
- Medium - Selection needs persistent state and a clear return to context.
- Low - Decorative mini-bars do not replace a sortable material ledger.

MRF manager reaction

- Good concept. I want one click from exception to evidence, not a lower-page exploration exercise.

Recommended changes

- Preserve behaviour. Surface selected detail near trigger or in a compact rail. Use a sortable ledger with scan count, confidence, contamination, weight/value, review status.

Sidebar and navigation

OVERALL DESIGN SCORE

4/10

What works

- Main workflow labels are short. Collapsed state is remembered.

Design problems

- Critical - Delayed sidebar class changes and transition suppression compound route instability.
- High - Collapsed icon-only navigation is not reliably understandable or accessible.
- High - Native anchors inside raw HTML weaken route continuity/state preservation.
- Medium - Workflow ordering does not show the next action.
- Medium - No explicit Dashboard label although Analytics performs that role.
- Low - Support/settings compete with the operating flow.

MRF manager reaction

- Where am I in the process, and what needs attention? The sidebar should answer both.

Recommended changes

- Repair route-shell ownership. Add accessible labels/tooltips. Visually separate Operate from Admin. Treat declarative Next navigation as a contained refactor.

Loading, empty, error, and success states

OVERALL DESIGN SCORE

3/10

What works

- Analytics shows genuine no-data, not fabricated output. Some retry handling exists.

Design problems

- Critical - Loading does not match final dimensions because route CSS fails and charts initialise late.
- High - Blank charts need a meaningful empty state and one next action.
- High - Analytics error retry performs a page reload rather than a scoped data retry.
- Medium - Colour alone carries too much status meaning.
- Medium - Success needs workflow continuity, not only a toast.
- Low - Global page fade makes the interface feel slower.

MRF manager reaction

- If loading, show what will appear. If failed, tell me whether data is unavailable or processing is ongoing.

Recommended changes

- Match skeletons to final layout. Reserve dimensions. Use inline retry with recovery guidance. Replace global fade with local content-state transitions.

Website-wide review

- Design consistency - moderate component consistency, weak route-scoped implementation; cards, radii, shadows, and overrides accumulate without a reliable shell.
- Navigation consistency - weak; sidebar state, native navigation, imperative body mutation, and custom-event initialisation are separate systems.
- Shared page shell - critical defect; React wrapper classes and body-scoped CSS conflict.
- Typography - IBM Plex suits operations, but chart/table labels are too small for long shifts.
- Spacing and components - generous but undisciplined; important areas crowd while containers retain empty space; card treatment is too uniform.
- Colour - green acts as brand, success, decoration, and neutral emphasis, so it loses operational meaning.
- Responsive/perceived performance - breakpoints exist, but shell and sidebar stability are weak; late scripts and broad initialisation make the application feel slow.
- Accessibility/credibility - focus styling exists, but collapsed navigation labels, chart alternatives, semantic statuses, and modal focus need work. Not yet credible for production MRF operations.

Fix now

- Make route styles follow the rendered page shell, not React-owned body.
- Scope initialisation to the current route; do not refresh/bind unrelated features.
- Reserve chart, KPI, queue, and upload-progress dimensions from first paint.
- Remove global body fade/translate from routing.
- Keep sidebar width stable and provide accessible labels in collapsed state.
- Preserve Analytics drill-through while repairing chart/render lifecycle.

Consider for redesign

- Reorder dashboard around exception-first decision-making.
- Reduce KPI/insight count and make warnings distinctive.
- Rebuild Result as a focused decision workbench.
- Redesign Log around unresolved exceptions and accountability.
- Establish semantic colour tokens for normal, review, contamination, hazardous, and neutral.
- Consolidate route shells into reusable React components and reduce globally loaded visual libraries.

Implementation gate

The navigation defects are confirmed. This report intentionally precedes implementation. The smallest safe repair is route-shell style ownership plus route-scoped lifecycle initialisation; it preserves upload processing, FastAPI, Supabase, Drive, review workflows, charts, and Analytics drill-through.